A Categorical Split in Havlík's Law: Liquid-Context Jers Never Vocalize

Marko Jevremović

Independent researcher, Belgrade, Serbia

marko@asemic.me

Abstract

This paper documents three independent lines of evidence that liquid-context jers in OCS were not real reduced vowels.

First, a categorical distributional split: strong-position jers in non-liquid environments vocalize to full vowels in 100% of cases across Czech ($n = 23$), Serbian ($n = 22$), and Slovak ($n = 23$)—68 out of 68 words—while strong-position jers adjacent to liquids vocalize in 0% of cases ($n = 27$). The seven Modern Czech words with /e/ near liquids (červ, černý, blecha, pleť') are secondary epenthesis of the 13th–15th centuries, documented in Old Czech as syllabic consonant forms (črv, črný, blcha, plt), and post-date the jer fall by centuries.

Second, a quality test: when Czech hard syllabic /l/ resolved, it yielded /u/ (žlutý, dlouhý), not /e/—the vowel that 100% of non-liquid jers produce. The quality tracks the liquid, not the jer.

Third, a quality diagnostic from Russian and Bulgarian: in both branches, the same front jer ь in strong position produces different vowels depending on the adjacent liquid. In Russian, ь yields /e/ before /r/ (верх, зерно, смерть) but /o/ before dark /l/ (волк, полный, жёлтый). In Bulgarian, ь yields /e/ in non-liquid contexts (ден, пес) but /ɤ/ (the back-jer reflex) before either liquid, in 12 of 12 etymologically eligible cases. If Havlík's law were operating, the conditioning would be the jer (front → /e/ regardless of environment). Instead, the conditioning is the liquid—the signature of liquid-conditioned epenthesis, the same mechanism that backs /e/ to /o/ before dark /l/ in East Slavic polnoglasie (melko → молоко).

We argue that liquid-context jers were notational conventions for already-syllabic consonants, and that Havlík's law applies only to genuine reduced vowels from PIE **i*/**u*. The apparent liquid exception disappears: there was never a vowel to vocalize.

1 Introduction

The fall of the jers (падение еров) is among the most consequential sound changes in Slavic historical phonology (Townsend & Janda 1996; Matasović 2014). Proto-Slavic had two reduced vowels, the back jer ъ and the front jer ь, which by the 10th–12th centuries were either strengthened to full vowels or lost entirely, depending on their position in the word. The governing regularity is Havlík's law: counting from the end of the word, jers in odd-numbered positions ("weak") are deleted, while jers in even-numbered positions ("strong") vocalize to full vowels—typically /e/ in Czech and Slovak, /o/ or /e/ in Russian (see Havlík 1889; Shevelov 1964; Bethin 1998).

The law is remarkably regular for jers in non-liquid environments. Proto-Slavic **sъnъ* 'sleep' yields Czech *sen*, **dьnь* 'day' yields *den*, **pьsъ* 'dog' yields *pes*—in each case the strong jer vocalizes to /e/ while the weak jer is deleted.

But for jers adjacent to liquids (/r/, /l/), the outcome is categorically different. Proto-Slavic **vьlkъ* 'wolf' yields Czech *vlk*—not **velk*. **sъmьrtь* 'death' yields *smrt*—not **smert*. **pьrstъ* 'finger' yields *prst*—not **perst*. The strong-position jer does not vocalize to /e/; instead, it disappears, and the liquid becomes a syllabic consonant.

This asymmetry is well known descriptively: any Czech grammar notes that the language has syllabic /r/ and /l/, and previous scholarship (Bethin 1998; Scheer 2003) has noted the structural distinctiveness of liquid-context jers. What has not been assembled, to the best of our knowledge, is a **systematic, cross-Slavic quantification** of the split together with the quality evidence from Russian. The categorical 68/0 pattern, combined with the /u/ test and the internal Russian /r/ vs. /l/ diagnostic, makes the simplest interpretation unavoidable: these were never vowels. In this paper we present this evidence and argue that the split constitutes grounds for a fundamental distinction between two types of graphemes written identically as ъ/ь in OCS:

1. **Real reduced vowels** (from PIE short **u*, **i*), which participate in Havlík's law normally;
2. **Notational conventions for syllabic consonants** (from PIE syllabic resonants **l̥*, **r̥*), which were never real vowels and therefore have nothing to vocalize.

This reanalysis eliminates the need for an ad hoc exception in Havlík's law for liquid contexts and simplifies the diachronic account of Slavic syllabic consonants.

2 Background

2.1 Havlík's law

Havlík's law (Havlík 1889) is typically formulated as follows. In a sequence of jers counted from the right edge of the word, the rightmost jer is weak (position 1), the next is strong (position 2), and so on, alternating. A full vowel resets the count. Weak jers are deleted; strong jers are vocalized to full vowels.[1]

The law operates with near-exceptionless regularity across Slavic for jers in non-liquid environments (see §3.1 below). Its status as a genuine phonological regularity

[1] The specific reflex varies by branch: /e/ in Czech and Slovak, /o/ (for ъ) or /e/ (for ь) in Russian, /a/ in Serbian, etc.

rather than a mere descriptive generalization is supported by modern experimental evidence: Gouskova & Becker (2013) showed that Russian speakers productively apply yer-like alternations to nonce words, with special constraints near sonorants.

2.2 PIE syllabic resonants and their Slavic reflexes

Proto-Indo-European had four sonorant consonants—**m*, **n*, **l*, **r*—that could function as syllable nuclei in zero-grade ablaut environments, written **m̥*, **n̥*, **l̥*, **r̥* (see Beekes 2011; Fortson 2010). Each IE daughter branch resolved these syllabic resonants differently, inserting a distinct epenthetic vowel: Sanskrit preserved the syllabic resonant (*r̥*), Greek inserted /a/, Latin /o/, Germanic /u/, Lithuanian /i/, and Slavic wrote jer + liquid (ь*r*, ъ*l*, etc.; Lunt 2001).[2] The standard reconstruction posits a Proto-Balto-Slavic intermediate stage **iR*/**uR* (Mayer 1991), with ∼73% showing **i* (reflected as front jer ь).

The question at issue is whether this intermediate PBS vowel was a "real" full short vowel that later reduced to a jer (the standard view), or whether Slavic preserved the syllabic resonant more or less directly from PIE, with the jer serving as a scribal device to represent the syllabic function of the consonant (the alternative we develop here).

2.3 Previous treatments of liquid-context jers

Scheer (2003, 2004) provides the most detailed structural analysis of Czech syllabic consonants within a Government Phonology (CVCV) framework. He distinguishes **syllabic consonants** (from Common Slavic **tьrt*, where the jer precedes the liquid) and **trapped consonants** (from CS **trьt*, where the jer follows). In his framework, the jer is lost and the liquid spreads into the vacated nuclear position. Scheer does not frame the vlk/sen difference as a problem for Havlík's law; in his framework, both outcomes are natural structural consequences of sonorant spreading.

Shevelov (1964) argued that Common Slavic jers were not a special "reduced" category but simply short vowels—the short members of the vowel system. This opens the theoretical space for liquid-context jers to have had a different phonetic realization (minimal or zero vocalic content) from non-liquid jers, even if they were written identically.

Birnbaum (1976) examined Slavic liquid diphthongs and their resolution across branches, noting the diversity of outcomes but without framing it as a Havlík's law problem. Shafirovich (2016) investigated "the possibility of existence of syllabic liquids in Common Slavic" by examining variation between ъ*r*/*t* and *r*ъ*t* spellings in manuscripts,

[2]The nasal syllabic resonants **m̥* and **n̥* followed a different path in Slavic, yielding nasal vowels (ѧ, ѫ) rather than jer + nasal sequences. They are therefore irrelevant to the Havlík's law asymmetry discussed here.

sometimes appearing in the same word without semantic distinction. This is the closest prior work to our argument, but it did not assemble the quantitative case we present here.

Isačenko (1970) proposed a revision of Havlík's law for East Slavic, arguing that the jer fall in Russian was governed by morphophonemic rather than purely phonological conditions. His revision focused on the interaction between jer vocalization and morphological boundaries, but did not isolate the liquid/non-liquid split as a categorical distinction. The present paper differs in identifying the asymmetry as a *phonological* fact about liquid adjacency, observable most cleanly in Czech, Serbian, and Slovak where the outcomes are transparent.

Bethin (1998) treats the liquid+jer interaction through the lens of prosodic structure, arguing that syllable reorganization preceded the fall of the jers and that liquid-adjacent jers occupied a distinct prosodic position. In a related vein, Yoo (2012), building on Bethin's (1992, 1993) autosegmental analysis of Late Common Slavic syllable structure, argues that the "tense jers" before jod in extreme eastern East Slavic dialects were not genuine jer vowels at all but reflect a different syllabification of the **ŭ*/ **ĭ*+jod sequence—a conclusion structurally parallel to ours for a different phonological environment.

None of these works assembles the quantitative cross-Slavic case presented here, nor combines it with the vowel-quality diagnostic from Russian. The present paper's contribution is not a new theoretical claim but a new **body of evidence**: the categorical 68/0 distribution, the /u/ test, and the /r/ vs. /l/ quality split, which together make the non-vocalic interpretation the simplest available account of the data.

3 The Czech data

Czech is the ideal primary test language for this investigation because it preserves both syllabic /r/ and /l/ as productive phonological categories, making the liquid-adjacent jer outcomes maximally transparent.

3.1 Category A: Strong jers not adjacent to liquids

Table 1 presents all identified cases where a strong-position jer in a non-liquid environment is attested in Proto-Slavic and has reflexes in Czech, Serbian, and Slovak—the three languages whose outcomes are most transparent for this investigation.[3]

The result is unambiguous: every strong-position jer in a non-liquid context vocalizes to a full vowel—/e/ in Czech and Slovak, /a/ in Serbian—across all three languages.

[3] Czech and Slovak preserve both syllabic /r/ and /l/ (Kenstowicz & Rubach 1987), making the liquid-adjacent contrast maximally visible. Serbian preserves syllabic /r/ and resolved syllabic /l/ to /u/, providing an independent test: if the jer had vocalized by Havlík's law, the outcome should be /a/ (the regular Serbian jer reflex), not /u/.

Table 1: Strong-position jers in non-liquid environments: cross-Slavic outcomes.

#	Proto-Slavic	Gloss	Czech	Serbian	Slovak
1	**sъnъ*	sleep	*sen*	*san*	*sen*
2	**dьnь*	day	*den*	*dan*	*deň*
3	**pьsъ*	dog	*pes*	*pas*	*pes*
4	**mъxъ*	moss	*mech*	*mah*	*mach*
5	**lьnъ*	flax	*len*	*lan*	*ľan*
6	**vъšь*	louse	*veš*	*vaš*	*voš*
7	**otьcь*	father	*otec*	*otac*	*otec*
8	**konьcь*	end	*konec*	*konac*	*koniec*
9	**kupьcь*	merchant	*kupec*	*kupac*	*kupec*
10	**šьvъ*	seam	*šev*	*šav*	*šev*
11	**lьstь*	cunning	*lest*	*last*[a]	*lesť*
12	**dъždžь*	rain	*déšť’*	*dažd*	*dážď’*
13	**ogňь*	fire	*oheň*	*oganj*	*oheň*
14	**dъska*	board	*deska*	*daska*	*doska*
15	**bъzъ*	elder tree	*bez*	*baz*	*baza*
16	**pьstrъ*	variegated	*pestrý*	—[b]	*pestrý*
17	**stьblo*	stalk	*stéblo*	*stablo*	*steblo*
18	**dьnьsь*	today	*dnes*	*danas*	*dnes*
19	**pьnь*	stump	*peň*	*panj*	*peň*
20	**zьtь*	son-in-law	*zeť*	*zet*[c]	*zať*
21	**tьnъkъ*	thin	*tenký*	*tanak*	*tenký*
22	**lьgъkъ*	light	*lehký*	*lak*	*ľahký*
23	**jьgъla*	needle	*jehla*	*igla*[d]	*ihla*
Strong jer vocalized:			**23/23**	**22/22**	**23/23**

[a]Archaic/literary; replaced by *lukavstvo* in colloquial Serbian. [b]Root lost in standard Serbian; replaced by *šaren*. [c]Serbian /e/ instead of expected /a/; the jer still vocalized, but with irregular quality. [d]Initial **jь-* regularly yields Serbian /i/ (cf. **jьmę* → *ime* ‘name’).

The vocalization rate is **100%** in each branch independently (Czech $n = 23$; Serbian $n = 22$, excluding one lost root; Slovak $n = 23$). Weak-position jers in the same words (**sъnъ*, **dьnь*) delete as predicted, confirming the standard strong/weak alternation.

Methodological note: The corpus was assembled by systematically extracting Proto-Slavic reconstructions containing jers from Derksen (2008), supplemented by Carlton (1991) and Shevelov (1964), and filtering for those with strong-position jers in non-liquid environments and transparent modern reflexes. The relevant statistical point is not the sample size but the **categorical nature** of the split—68/68 vocalized jers across three independent branches vs. 0/27 in liquid-adjacent environments—which no reasonable alternative model predicts. Even if additional examples exist, a single liquid-adjacent strong jer that vocalized to /e/ would suffice to falsify the categorical claim; none has been found.

3.2 Category B: Strong jers adjacent to liquids

Table 2 presents strong-position jers adjacent to /r/ (from PIE syllabic **r̥*), and Table 3 those adjacent to /l/ (from PIE syllabic **l̥*).

Table 2: Strong-position jers adjacent to /r/: Czech outcomes.

#	Proto-Slavic	Gloss	PIE	Czech	→ /e/?	Outcome
1	**sъmьrtь*	death	**mr̥tis*	*smrt*	no	syll. *r*
2	**mьrtvъ*	dead	**mr̥twós*	*mrtvý*	no	syll. *r*
3	**vьrxъ*	top	**wr̥s-*	*vrch*	no	syll. *r*
4	**tъrgъ*	market	—	*trh*	no	syll. *r*
5	**kъrkъ*	neck	**(s)ker-*	*krk*	no	syll. *r*
6	**pьrstъ*	finger	**pr̥sth*$_2$	*prst*	no	syll. *r*
7	**tьrnъ*	thorn	**tr̥nós*	*trn*	no	syll. *r*
8	**pьrvъ*	first	**pr̥h*$_2$*wós*	*prvý*	no	syll. *r*
9	**gъrdlo*	throat	**g*^{w}*r̥h*$_3$*d*^{h}*lom*	*hrdlo*	no	syll. *r*
10	**zьrno*	grain	**ǵr̥h*$_2$*nóm*	*zrno*	no	syll. *r*
11	**sьrdьce*	heart	**ḱr̥d-*	*srdce*	no	syll. *r*
12	**čьrvь*	worm	**k*^{w}*r̥mis*	*červ*	SEC.	OldCz. *črv*
13	**čьrnъ*	black	**kr̥snós*	*černý*	SEC.	OldCz. *črný*
14	**čьrvenъ*	red	← **čьrvь*	*červený*	SEC.	OldCz. *črvený*
15	**čьrstvъ*	stale	—	*čerstvý*	SEC.	OldCz. *črstvý*
16	**bьrvьno*	beam	**b*^{h}*r̥w-*	*břevno*	SEC.	epenthesis
17	**čьrta*	line	**(s)kr̥t-*	*črt* / *čára*	mixed	*črt* = syll. *r*

Results for Category B:

- Syllabic consonant preserved (no vocalization at all): **16 words**
- Hard syllabic *l* → /u/ (not /e/): **4 words**

Table 3: Strong-position jers adjacent to /l/: Czech outcomes.

#	Proto-Slavic	Gloss	PIE	Czech	→ /e/?	Outcome
18	**vьlkъ*	wolf	**wl̥kʷos*	*vlk*	no	syll. *l*
19	**pьlnъ*	full	$*pl̥h_1nos$	*plný*	no	syll. *l*
20	**vьlna*	wool	$*h_2wl̥h_1neh_2$	*vlna*	no	syll. *l*
21	**slьza*	tear	—	*slza*	no	syll. *l*
22	**žьltъ*	yellow	$*ǵ^hl̥h_3tos$	*žlutý*	no → /u/	hard l → lu
23	**dьlgъ*	long	$*dl̥h_1g^hos$	*dlouhý*	no → /u/	hard l → lu → ou
24	**dъlgъ*	debt	$*d^helg^h$-	*dluh*	no → /u/	hard l → lu
25	**tъlstъ*	thick	—	*tlustý*	no → /u/	hard l → lu
26	**blъxa*	flea	—	*blecha*	SEC.	OldCz. *blcha*
27	**plъtь*	flesh	$*pleth_2$-	*pleť*	SEC.	OldCz. *plt*

- Secondary epenthesis (13th–15th c., from Old Czech syllabic consonant): **7 words**
- Direct vocalization to /e/ by Havlík's law: **0 words**

The vocalization rate for liquid-adjacent strong jers is **0%** ($n = 27$).

3.3 The contingency table

Table 4: Havlík's law outcomes by environment (Czech, strong-position jers only).

	Vocalized to /e/	Did not vocalize to /e/
Non-liquid environment	23	0
Liquid-adjacent environment	0	27

Fisher's exact test on the 2×2 table yields $p < 10^{-14}$. The asymmetry is not a statistical tendency; it is categorical. The same split is replicated independently in Serbian (22/22 vs. 0) and Slovak (23/23 vs. 0).

4 The secondary epenthesis argument

Seven Category B words show /e/ in Modern Czech: *červ* 'worm', *černý* 'black', *červený* 'red', *čerstvý* 'stale', *břevno* 'beam', *blecha* 'flea', and *pleť* 'complexion'. A reviewer might object that these represent Havlík's law vocalization and thus constitute counterexamples. Three lines of evidence demonstrate that they do not.

4.1 Old Czech attestation

Historical texts from the 13th–14th centuries attest Old Czech forms **without** /e/: *črv* (not *červ*), *črný* (not *černý*), *črvený* (not *červený*), *črstvý* (not *čerstvý*), *brvno* (not *břevno*), *blcha* (not *blecha*), *plt* (not *plet'*). These forms are attested in 13th–14th century sources including the *Dalimil Chronicle* (ca. 1314) and other early Old Czech texts (see Lamprecht 1987: 104–112; Komárek 1969 for a systematic overview). The syllabic consonant stage is directly documented in the written record. The /e/ appeared later.

4.2 Chronological separation

Havlík's law operated during the 10th–12th centuries—the period of the "fall of the jers" across Slavic. The secondary epenthesis that produced *črv* → *červ* and *blcha* → *blecha* occurred in the **13th–15th centuries**, well after the jer fall was complete. These are two distinct sound changes separated by centuries. The /e/ in *černý* was never a jer; it is a later innovation to resolve consonant clusters that had become too complex.

4.3 The /u/ test

When hard syllabic /l/ resolved in Czech, the epenthetic vowel was /u/, not /e/:

**žьltъ*	→ *žlutý*	(not **žletý*)
**dьlgъ*	→ *dlouhý*	(via *dlúhý*, not **dlehý*)
**dъlgъ*	→ *dluh*	(not **dleh*)
**tъlstъ*	→ *tlustý*	(not **tlestý*)

If Havlík's law had vocalized these strong jers, the output should have been /e/—as it is in 100% of Category A cases. The fact that /u/ appears instead proves that a different process was at work: resolution of a syllabic consonant, not vocalization of a jer. The vowel quality tracks the phonetic character of the liquid (back /u/ for velarized /l/), not the quality predicted by jer vocalization. This is the same generalization that the Russian data will confirm independently (§5.2): in both Czech and Russian, the vowel quality at liquid-adjacent positions is conditioned by the *liquid*, not by the jer.

5 Cross-Slavic evidence

The asymmetry documented in Czech is not a Czech-internal peculiarity. Other Slavic branches show parallel patterns, each in their own way.

5.1 Serbian

Serbian vocalizes non-liquid strong jers to /a/ consistently: **sъnъ* → *san*, **dьnь* → *dan*, **pьsъ* → *pas*. For liquid-adjacent jers, the picture is different:

- **Syllabic /r/ is preserved** across a broad inventory: *smrt* 'death', *prst* 'finger', *crn* 'black', *trn* 'thorn', *trg* 'market', *vrh* 'top', *krv* 'blood', *srp* 'sickle', *grb* 'hump', *brz* 'fast', *grlo* 'throat', *zrno* 'grain', *srce* 'heart', *drvo* 'wood', *tvrd* 'hard', *crv* 'worm', *prvi* 'first'—at least 20 common words, all from PIE syllabic **r̥* or Proto-Slavic ь*r* positions.

- **Syllabic /l/ resolved to /u/**: *vuk* 'wolf' (< *vlk*), *vuna* 'wool' (< *vlna*), *pun* 'full' (< *pln*), *žut* 'yellow' (< ∗*žlt*).

The vowel inserted for **l̥* positions is /u/, not /a/. If Havlík's law had vocalized these jers, we would expect /a/—the same reflex as all Category A jers. The /u/ points to a different process: resolution of a syllabic consonant.

A further piece of evidence comes from the 19th-century Serbian spelling reform. When Vuk Karadžić removed jers from the orthography, words like трьнъ became *trn* with **no change in pronunciation**. The /r/ did not "become" syllabic at that point—it already was. The jer had been an orthographic fossil for centuries.

The **asymmetry between /r/ and /l/** is itself informative. Under the standard model, both positions contained "the same jer": why would they behave differently? Under our model, both were syllabic consonants; syllabic /r/ was more typologically stable than syllabic /l/ and survived longer, while /l/ independently resolved to /u/ (completed by ca. 1400, based on manuscript evidence).

5.2 Russian: The polnoglasie connection

Russian presents an initially challenging case for our argument. East Slavic vocalized **all** jers in liquid contexts to full vowels: **vъlkъ* → волк [volk], **sъmьrtь* → смерть [$sm^j ert^j$], **pьrstъ* → перст [$p^j erst$]. This appears to support the standard model. However, closer examination of the vowel quality reveals that the Russian data *support* rather than contradict our proposal.

5.2.1 The vowel quality diagnostic

The key move is not simply that liquid-context jers "fail to vocalize" in Russian—they appear to vocalize. The decisive evidence is the **quality** of the resulting vowel.

Under Havlík's law, jer quality determines vowel quality: front ь → /e/, back ъ → /o/. In non-liquid contexts, Russian follows this consistently (**dьnь* → день /e/, **sъnъ* → сон /o/). The same prediction holds for ь before /r/. But for ь before /l/, it fails—and the failure is conditioned by the **liquid**, not the jer:

The same jer (ь, front), the same strong position, the same branch—but two different liquids produce two different vowel qualities. If Havlík's law were operating, the conditioning would be the **jer** (front → /e/ regardless of environment). Instead, the

Table 5: Front jer ь in Russian: /r/ contexts vs. /l/ contexts.

Proto-Slavic	Jer	Liquid	Havlík predicts	Russian	Vowel	Match?
**vьrxъ*	ь (front)	/r/	/e/	верх	/e/	yes
**pьrstъ*	ь (front)	/r/	/e/	перст	/e/	yes
**sъmьrtь*	ь (front)	/r/	/e/	смерть	/e/	yes
**zьrno*	ь (front)	/r/	/e/	зерно	/e/	yes
**vьlkъ*	ь (front)	/l/	/e/	волк	/o/	**no**
**pьlnъ*	ь (front)	/l/	/e/	полный	/o/	**no**
**žьltъ*	ь (front)	/l/	/e/	жёлтый	/o/	**no**

conditioning is the **liquid**: /r/ permits /e/; dark /l/ forces backing to /o/. This is the diagnostic signature of independent epenthesis into a syllabic resonant, where the inserted vowel's quality is determined by the surrounding consonants, not by any inherited jer quality (Townsend & Janda 1996: 84–86).

5.2.2 The liquid-conditioned backing mechanism

The /r/ vs. /l/ contrast in Table 5 is not an isolated anomaly. It is the same phonological operation that operates in East Slavic polnoglasie (pleophony; Shevelov 1964: 421–446)—though we are not claiming these are the same historical process. Polnoglasie proper concerns TORT/TOLT liquid diphthongs; liquid-jer resolution is a distinct diachronic event. What they share is a **phonological mechanism**: epenthesis of a vowel whose quality is conditioned by the adjacent liquid, with velarized dark /l/ forcing /o/ and non-velarized /r/ permitting /e/.

East Slavic polnoglasie illustrates the mechanism clearly:

**gordъ* (TORT)	→ город (ToroT)	vowel copy after liquid
**golva* (TolT)	→ голова (ToloT)	vowel copy after liquid
**bergъ* (TerT)	→ берег (TereT)	vowel copy after liquid
**melko* (TelT)	→ молоко (ToloT)	/e/ backed to /o/ before dark /l/

The **melko* case is critical: a real, uncontroversial /e/ is backed to /o/ before dark /l/. The same operation produces the same output in волк, полный, жёлтый—on what the standard model calls a "vocalized front jer." The simpler hypothesis is that in both cases the mechanism is identical: epenthesis of a vowel whose quality is determined by the liquid, not by any inherited jer quality.

Context	Input	East Slavic output	Conditioning
Polnoglasie, /r/	T**e**rT	T**e**r**e**T	/r/ → /e/ preserved
Polnoglasie, /l/	T**e**lT	T**olo**T	dark /l/ → /e/ backed to /o/
Liquid-jer, /r/	T**ь**rT	T**e**rT	/r/ → /e/
Liquid-jer, /l/	T**ь**lT	T**o**lT	dark /l/ → /o/

In all four cases, the vowel quality is determined by the liquid. The jer column is irrelevant to the outcome.

5.2.3 The geographic isogloss

The geographic split between South Slavic and East Slavic is **identical** for both TORT and TьRT words:

	South Slavic	East Slavic
**gordъ* (TORT)	градъ (metathesis)	город (polnoglasie)
**golva* (TolT)	глава (metathesis)	голова (polnoglasie)
**vъlkъ* (TъlT)	вльκъ > Serb. *vuk*	волк (/o/ epenthesis)
**sъmьrtь* (TьrT)	Serb. *smrt* (syll. *r*)	смерть (/e/ epenthesis)
**pьrstъ* (TьrT)	Serb. *prst* (syll. *r*)	перст (/e/ epenthesis)

In every case, South Slavic preserves or reconfigures the consonant cluster, while East Slavic breaks it up with a full vowel. This is the same phonotactic constraint operating on two successive waves of CRC clusters: the TORT sequences (8th–9th c.) and the TьRT sequences exposed as CRC clusters after the jer fall (10th–12th c.).

5.2.4 Reinterpretation

We propose that what is traditionally described as "vocalization of liquid-context jers" in East Slavic is better analyzed as **polnoglasie applied to syllabic consonants**. East Slavic had a productive phonotactic constraint against CRC sequences; when the jer fall exposed syllabic consonants as CRC clusters, the same epenthetic mechanism that had resolved TORT sequences was applied to Tl̥T and Tr̥T sequences. The vowel quality was determined by the liquid (backing before /l/, preservation before /r/), not by a jer that had already ceased to exist.

Under this analysis, Russian does not constitute evidence *for* the standard model. The East Slavic data are compatible with our proposal: the jers in liquid contexts were non-vocalic, and their apparent "vocalization" was in fact independent epenthesis—the same process, by a different name, that every other IE branch also applied to PIE syllabic resonants.

Further support comes from the Old Novgorod dialect (Zaliznyak 2004), which shows TъRъT / TьRьT patterns—a **copy** of the jer inserted after the liquid, structurally identical to how polnoglasie copies the full vowel. In Novgorod, the two jers apparently *neutralized* in this position, and the resulting vowel quality was conditioned by the consonantal environment rather than the original jer color.

5.3 Bulgarian: a fourth categorical branch

Bulgarian provides a fourth independent confirmation of the categorical asymmetry. Non-liquid front jer ь regularly yields /e/ (ден < **dьnь*, пес < **pьsъ*). If Havlík's law had applied to liquid-adjacent jers, front jer ь before a liquid should likewise yield /e/. It does not:

Table 6: Bulgarian: front jer ь in non-liquid vs. liquid-adjacent contexts.

Proto-Slavic	Jer	Gloss	Havlík predicts	Bulgarian	Actual	Match?
**dьnь*	ь	day	/e/	ден	/e/	yes (non-liquid)
**pьsъ*	ь	dog	/e/	пес	/e/	yes (non-liquid)
**vьlkъ*	ь	wolf	/e/	вълк	/ɤ/	**no**
**pьlnъ*	ь	full	/e/	пълен	/ɤ/	**no**
**žьltъ*	ь	yellow	/e/	жълт	/ɤ/	**no**
**vьlna*	ь	wool	/e/	вълна	/ɤ/	**no**
**pьrstъ*	ь	finger	/e/	пръст	/ɤ/	**no**
**tьrnъ*	ь	thorn	/e/	трън	/ɤ/	**no**
**vьrxъ*	ь	top	/e/	връх	/ɤ/	**no**
**zьrno*	ь	grain	/e/	зърно	/ɤ/	**no**
**sьrdьce*	ь	heart	/e/	сърце	/ɤ/	**no**
**mьrtvъ*	ь	dead	/e/	мъртъв	/ɤ/	**no**

All ten liquid-adjacent front-jer words yield /ɤ/, not the /e/ predicted by Havlík's law. The vowel quality (a back vowel, identical to the regular reflex of back jer ъ) is determined by the consonantal environment, not by the jer color—paralleling the Russian /r/ vs. /l/ diagnostic of §5.2, but extended across both liquids.

Two apparent counterexamples are secondary developments. Standard Modern Bulgarian черен 'black' (from **čьrnъ*) and червей 'worm' (from **čьrvь*) show /e/ rather than the expected /ɤ/. However, Bulgarian dialects preserve цьрн, чьрн, црьн 'black' and цьрв, чьрв, црьв 'worm' with /ɤ/, exactly as the pattern predicts.[4] These cases parallel the Czech secondary epenthesis documented in §4 (*črv* → *červ*, *črný* → *černý*, both 13th–15th c. innovations): in both branches, original syllabic-consonant forms were later resolved by epenthesis of /e/ in the same etymological cluster (Modern Slavic *čьr-). Counting the dialectal/older forms with /ɤ/, Bulgarian shows the pattern in 12 of 12 etymologically eligible cases.

Table 7: Cross-Slavic treatment of liquid-adjacent jers vs. non-liquid jers.

Branch	Non-liquid jer →	Liquid jer →	Same process?
Czech/Slovak	/e/	syllabic C	**no**
Serbian	/a/	syll. *r* (kept) / /u/ (*l*)	**no**
Russian	/o/ or /e/	/o/ or /e/ (liquid-conditioned)	**no** (see §5.2)
Bulgarian	/e/ or /ɤ/	/ɤ/ (liquid-conditioned)	**no** (see §5.3)

5.4 The cross-Slavic summary

In all four branches examined in detail, liquid-adjacent jers **behave differently** from non-liquid jers.[5] For Czech, Serbian, Slovak, and Bulgarian the distinction is categorical; for Russian, the vowel quality in liquid-adjacent positions follows liquid-conditioned epenthesis patterns rather than standard jer vocalization (§5.2). The convergence of four independent branches on the same categorical asymmetry is difficult to attribute to any single later innovation; it is most naturally explained by the hypothesis that liquid-context jers were not real vowels in the input to Havlík's law.

6 Engagement with the standard account

6.1 The implicit exception

The standard account of Havlík's law does not contain an explicit exception clause for liquid contexts. Instead, textbooks present jer vocalization and syllabic consonant formation as complementary outcomes of the same process, without flagging the logical tension. A representative formulation might read: "Strong jers vocalized; in liquid environments, the liquid became syllabic." But this formulation masks a genuine asymmetry: *why* does the liquid "become syllabic" instead of the jer vocalizing, as it does in every non-liquid context?

The standard answer, insofar as one is given, relies on rule ordering: the jer weakens, the liquid—being the most sonorous adjacent segment—fills the vacated nuclear position. But this is precisely the kind of post hoc explanation that our alternative renders unnecessary.

[4] Old Church Slavonic forms чрьнъ, чрьвь (with no jer vocalization) and modern dialectal/toponymic цьрн- are attested in the Старобългарски речник (Old Bulgarian Dictionary, Vol. II). The toponymic evidence is particularly clear: place names such as Цьрниниви, Цьрна гора, Цьрна река, Цьрна трава, and Цьрноок preserve /ɤ/ across Bulgarian dialects, while the standard /e/ forms appear to be later developments parallel to Czech *črv* → *červ* (§4).

[5] Polish also shows distinct treatment of liquid-adjacent jers, with variable outcomes (/i/, /e/, /a/) contrasting with uniform /e/ for non-liquid jers (Rubach 1993). However, the Polish situation is complicated by dialect variation and morphological conditioning, and we defer a full treatment to future work.

6.2 Scheer's structural analysis

Scheer's (2003, 2004) CVCV analysis of Czech syllabic consonants provides the most detailed structural account of liquid-jer interactions in the literature. In his Government Phonology framework, syllable structure is represented as a strict alternation of consonantal (C) and vocalic (V) positions. A syllabic consonant is analyzed as a liquid that spreads into an adjacent empty vocalic position, acquiring nuclear properties (stress-bearing, mora-contributing). Scheer distinguishes two structural types in Czech:

1. **Syllabic consonants** (from CS **tьrt*, jer before liquid): the jer is lost, leaving an empty nucleus which the liquid fills through leftward spreading. Example: **vьlkъ* → *vlk*, where /l/ spreads into the nuclear position originally marked by ь.

2. **Trapped consonants** (from CS **trьt*, jer after liquid): the jer is lost, and the liquid is "trapped" between two consonants, becoming syllabic through government relations. Example: **bьrati* → *brát*, where /r/ is trapped and syllabifies.

Scheer's analysis is **structurally compatible with both the standard and alternative models**. The key point is that the synchronic phonological structure—a liquid associated with a nuclear position—is *identical* under both interpretations:

- **Standard model:** The nuclear position was occupied by a real reduced vowel (PBS **i*), which later deleted, leaving an empty nucleus that the liquid filled.
- **Alternative model (ours):** The nuclear position was *always* empty (or minimally vocalic), with the liquid functioning as the nucleus from the start; the jer was a scribal convention marking this syllabic function.

In both cases, the *outcome* is a liquid in a nuclear position. The difference is purely diachronic: was there an intermediate stage with a full vowel, or was the liquid quasi-syllabic throughout?

Scheer's framework does not adjudicate this question because it is designed to model synchronic phonological structure, not historical phonology. His analysis takes the OCS spelling (jer + liquid) as a starting point and derives the Modern Czech outcome (syllabic consonant) through independently motivated phonological operations (nuclear deletion, segmental spreading, government). The framework is neutral on whether the OCS jer notation represented a real vowel or a notational device.

What Scheer's analysis contributes to our argument: Two key insights from Scheer support the broader claim that OCS jer notation was functionally heterogeneous:

1. **Structural heterogeneity of jers.** Scheer shows that syllabic consonants (**tьrt*) and trapped consonants (**trьt*) are **structurally distinct** categories in Modern

Czech, even though both were written with jer in OCS. This demonstrates that OCS jer notation did not represent a uniform phonological category—what looks identical orthographically (ь) corresponds to different underlying structures. Our proposal extends this insight: not only do jers in different *positions* (before vs. after liquids) correspond to different structures, but jers in liquid contexts vs. non-liquid contexts correspond to fundamentally different *types* of segment (notational vs. real vowel).

2. **Independent evidence for empty nuclei.** Scheer's framework requires that certain positions contain **empty nuclei**—vocalic positions in the syllable structure that have no segmental content. Under the standard model, these empty nuclei arise *after* the jer is deleted. Under our model, they were empty (or quasi-empty) *from the start*. Either way, the existence of structurally projected but phonetically empty vocalic positions is independently required by Modern Czech phonotactics. Our proposal simply pushes the origin of these empty nuclei earlier in the historical timeline.

What our data adds: Scheer's analysis does not explain—or attempt to explain—the categorical 100% vs. 0% vocalization asymmetry documented in §3. His framework derives the *outcome* (syllabic consonants) from the *input* (OCS jer + liquid), but it is silent on why Havlík's law vocalization operates categorically differently in liquid vs. non-liquid environments. In Scheer's terms, one could ask: why does the strong-position nuclear slot vocalize to /e/ in **sъnъ* → *sen* but remain empty (to be filled by liquid spreading) in **vьlkъ* → *vlk*? The CVCV framework has no mechanism to predict this split.

Our answer is historical: the nuclear slot in **sъnъ* was occupied by a real vowel (PIE **u*), which vocalizes under Havlík's law; the nuclear slot in **vьlkъ* was never occupied by a real vowel (PIE **l̥* was quasi-syllabic), so there was nothing to vocalize. The asymmetry arises from a difference in *etymological input*, not synchronic phonological structure.

A note on trapped consonants is in order. Our argument has focused on the **tьrt* type (jer before liquid), where the strong-position jer fails to vocalize. What about the **trьt* type (jer after liquid), where Scheer posits a structurally distinct "trapped" outcome? Under our model, both jer positions are notational: neither the jer in **vьlkъ* nor the jer in **bьrati* represents a real reduced vowel. The prediction is the same in both cases—the jer should not participate in Havlík's law—and the Czech outcomes confirm this:

Proto-Slavic	Type	Czech	Outcome	Jer vocalized?
**bьrati*	**trьt*	*brát*	trapped *r*	no (á < root vowel)
**krьstъ*	**trьt*	*kříž*	trapped *r*	no
**drьva*	**trьt*	*dříví*	trapped *r*	no
**blьstěti*	**tlьt*	*blýskat*	trapped *l*	no

In no case does the jer in a trapped-consonant position vocalize to /e/. We note, however, that these data are *neutral* between the two models: Scheer's standard CVCV analysis independently predicts that trapped-consonant jers do not vocalize (the liquid spreads into the vacated nuclear position before vocalization can apply). The trapped-consonant cases are therefore compatible with our proposal but do not constitute independent evidence for it. The distinguishing evidence comes from the **tьrt* type (§3), where the standard model predicts vocalization and our model does not. Scheer's structural distinction between syllabic and trapped consonants remains valid as a synchronic description; our proposal adds the diachronic claim that neither type ever contained a real vowel at the jer position.

In summary, Scheer's structural analysis and our diachronic proposal are complementary. Scheer provides the synchronic phonological machinery to derive syllabic consonants from underlying representations; we provide the historical explanation for why the underlying representations differ categorically between liquid and non-liquid environments.

6.3 Contrast with Bethin (1998)

Bethin (1998) argues that pre-liquid jers in roots like **vьlkъ* indicate the syllabicity of the adjacent liquid, which is preserved after the jer's deletion. This position is closely related to ours, and it is worth being explicit about where the two accounts agree and where they differ.

	Bethin (1998)	This paper
Was there a vowel at the jer position in OCS?	Yes: a reduced vowel, subsequently deleted. Syllabicity of the liquid is a consequence of deletion.	No: the jer was a notational convention for an already-syllabic consonant. There was no vowel to delete.
Why doesn't the jer vocalize?	The liquid's prosodic properties preempt vocalization (syllable reorganization precedes the jer fall).	There is no vowel to vocalize.
What does the Russian /o/ in волк show?	Not directly addressed.	Front jer before dark /l/ gives /o/ not /e/ — the vowel quality is conditioned by the liquid, ruling out Havlík vocalization.

The critical empirical difference is the Russian quality diagnostic (Table 5). Under Bethin's account, the jer in **vьlkъ* is a real front vowel undergoing syllable reorganization; if it were vocalized by Havlík's law, it should yield /e/. The fact that Russian yields /o/—and that this /o/ is conditioned by the dark /l/, not by the jer—is not predicted by a model in which a real front vowel was present at that position. Under our account, no jer vowel was ever present; the /o/ reflects liquid-conditioned epenthesis operating on a syllabic consonant, the same mechanism that backs /e/ to /o/ in **melko* → молоко.

A related formal account is Scheer & Ziková (2010), who propose that the Havlík alternation pattern arises from directional (right-to-left) application of Lower (Lightner 1965), with cyclic vs. non-cyclic affix structure determining the pattern. Their synchronic analysis is orthogonal to our diachronic argument: they model the alternation pattern for genuine reduced vowels; we document why one class of inputs—liquid-adjacent jers—sits entirely outside that alternation. The categorical 68/0 split is invisible under any account that treats liquid-context and non-liquid-context jers as phonologically equivalent inputs to the same rule.

6.4 The ъ/ь distinction

One argument for the standard model is that the front/back jer distinction in liquid contexts (ь*r* vs. ъ*r*) implies a vocalic element with quality. A purely syllabic consonant has no front/back dimension.

A systematic survey of 74 Proto-Slavic words with jer adjacent to a liquid (*r* or *l*) substantially weakens this objection. The jer color in these positions turns out to be **etymologically determined**, with near-categorical accuracy (Table 8): front jer ь correlates almost perfectly with PIE syllabic resonant origin, while back jer ъ correlates perfectly with non-syllabic-resonant sources (loanwords, laryngeal sequences, substrate words). The minimal pair **dьlgъ* 'long' (from PIE $*d\underset{\circ}{l}h_1g^h os$, front jer) vs. **dъlgъ* 'debt' (from Gothic, back jer) epitomizes the pattern: same consonant skeleton, opposite jer color, opposite etymological origin.

Table 8: Jer color and etymological source in liquid contexts ($n = 74$).

	PIE $*\underset{\circ}{r}/*\underset{\circ}{l}$	Other source	Total
ь + liquid	**35** (94.6%)	2 (5.4%)	37
ъ + liquid	0 (0%)	**37** (100%)	37

Representative examples illustrate the pattern:

Proto-Slavic	Jer	Gloss	Source	PIE syll. resonant?
**vьlkъ*	ь	wolf	PIE $*w\underset{\circ}{l}k^w os$	**yes**
**pьrstъ*	ь	finger	PIE $*p\underset{\circ}{r}sth_2$	**yes**
**čьrnъ*	ь	black	PIE $*k\underset{\circ}{r}sn\acute{o}s$	**yes**
**zьrno*	ь	grain	PIE $*\acute{g}\underset{\circ}{r}h_2n\acute{o}m$	**yes**
**dьlgъ*	ь	long	PIE $*d\underset{\circ}{l}h_1g^h os$	**yes**
**vьlna*	ь	wool	PIE $*h_2w\underset{\circ}{l}h_1neh_2$	**yes**
**dъlgъ*	ъ	debt	Gothic *dulgs*	no (loanword)
**xъlmъ*	ъ	hill	PGmc **hulmaz*	no (loanword)
**gъrdlo*	ъ	throat	PIE $*g^w\underset{\circ}{r}h_3d^h lom$	no (laryngeal)
**kъrmiti*	ъ	to feed	PIE $*k^w\underset{\circ}{r}Hm$-	no (laryngeal)
**tъrgъ*	ъ	market	substrate	no (substrate)
**tъlstъ*	ъ	thick	unknown	no (unknown)

This distribution shows that jer color in liquid contexts encodes inherited PBS-level etymological information (corresponding to the PBS **iR*/**uR* split; Mayer 1991), not independent Slavic vocalic quality—exactly what one would expect of a notational system preserving traditional spelling distinctions from an earlier stage, not of a phonologically active vowel whose quality is determined by its synchronic environment.[6]

6.5 The Balto-Slavic agreement

Baltic and Slavic agree on front (**i*) vs. back (**u*) quality in the same liquid-context words (∼73% **i*; Mayer 1991), suggesting a shared PBS innovation rather than independent development. This is genuine evidence for a PBS stage with *some* vocalic quality in these positions. We do not dismiss this evidence.

[6] The full 74-word corpus, conditioning environments, and diagnostic applications of the jer-color pattern are developed in a separate study.

However, the agreement is equally compatible with a PBS stage where the resonant was quasi-syllabic with incipient vocalic coloring—not a full vowel. Baltic developed this coloring into a full vowel **i* (Lithuanian *vilkas*, *vilna*, *pilnas*); Slavic preserved the syllabic consonant with the inherited coloring encoded in the ъ/ь spelling (see §6.4). The agreement on front/back quality requires a shared starting point, but that starting point need not have been a full short vowel.

The two models make a concrete, distinguishable prediction. If the PBS stage contained a real short vowel, that vowel should participate in Havlík's law like all other real reduced vowels—that is, it should vocalize in strong position. The data in §3 show that it categorically does not. This is the empirical test that distinguishes "quasi-syllabic consonant with incipient coloring" from "real short vowel": a real vowel vocalizes; liquid-context jers never do.

A clarification of terminology is warranted here. When we call liquid-context jers "notational," we do not mean that they encoded nothing real. The ъ/ь distinction in these positions faithfully recorded a genuine phonetic property—the front or back coloring of the syllabic resonant, inherited from PBS. What we mean is that this coloring was a *secondary property of the consonant*, not an independent vocalic segment. The jer grapheme represented the consonant's coloring, much as a diacritic represents a secondary articulation, rather than a separate vowel occupying its own position in the syllable. The critical difference is phonological status: a full short vowel participates in vowel-specific processes (Havlík's law vocalization); a colored consonant does not. The 68/68 vs. 0/27 split is the empirical reflex of this distinction.

Furthermore, the Balto-Slavic agreement on quality distribution is itself informative for our argument. Mayer's (1991) comprehensive study of 215 PBS lexical items shows that the front (**i*) reflex accounts for ∼73% of cases, back (**u*) for ∼17%, and both for ∼10%. If the PBS vowel had been a robust, phonologically active short vowel, we would expect the conditioning of its front/back quality to be well understood. In fact, over a century of scholarship has failed to establish the conditioning definitively (proposals by Vaillant 1950, Kuryłowicz 1956, Endzelīns 1971, and Stang 1957 all capture only a subset of cases). This opacity is more consistent with a quasi-syllabic resonant whose incipient coloring was phonetically variable than with a firmly established short vowel.

7 The proposal

7.1 Two types of jer

We propose that the OCS graphemes ъ and ь served at least three distinct functions, two of which are phonologically genuine and one of which is notational:

A fourth function—loanword epenthesis—is included for completeness: words like

Table 9: Three functions of the jer graphemes.

Function	Example	PIE source	Real vowel?
Reduced vowel (< PIE **u*)	сънъ 'sleep'	**supnos*	**yes**
Reduced vowel (< PIE **i*)	дьнь 'day'	**dyew-*	**yes**
Syllabic C notation	влькъ 'wolf'	**wl̥kʷos*	**no**
Loanword epenthesis	стькло 'glass'	Gmc **stikls*	**no**

стькло 'glass' (from Germanic **stikls*) acquired a jer in Slavic to break up consonant clusters foreign to Slavic phonotactics. These epenthetic jers, like syllabic-consonant jers, were not inherited reduced vowels and similarly fall outside the domain of Havlík's law.[7]

Havlík's law applies to the first two categories: genuine reduced vowels that alternate between vocalization (strong position) and deletion (weak position). It does not apply to the third or fourth categories, because there is no inherited vowel to vocalize. The 100% vs. 0% split in the Czech data (Tables 1–3) reflects this categorical distinction.

An alternative interpretation should be considered and rejected. One might propose that liquid-context jers *were* real vowels but ultra-short ones—phonetically present but already below the threshold for Havlík's law vocalization. On this view, the 0/27 result reflects phonological inertness, not absence. Five arguments weigh against this alternative:

1. **Parsimony.** The "inert vowel" hypothesis requires positing a special phonological category—vowels too short to vocalize yet real enough to carry front/back quality—that has no independent motivation in Slavic. We would need to explain why *only* liquid-adjacent jers fell into this category while all other jers, including those in complex clusters (**pьstrъ → pestrý*, **dъždžь → déšť*), vocalized normally.

2. **The /u/ test.** When hard syllabic /l/ resolved in Czech, it produced /u/ (*žlutý*, *dlouhý*)—the same outcome regardless of whether a jer was once present. If liquid-context jers had been real vowels, however short, we would expect some interaction with the resolution process. Instead, the vowel quality tracks the liquid consonant, exactly as in positions where no vowel was ever posited.

3. **The polnoglasie parallel.** Russian resolves liquid-context jers by the same epenthetic mechanism it applies to TORT sequences (§5.2), where no jer was ever present. If liquid-context jers had been real vowels, there would be no reason for Russian to treat them identically to vowelless consonant clusters.

[7] Compare Czech *sklo* (jer deleted, no vocalization) vs. *sen* < **sъnъ* (jer vocalized). The loanword jer in **stьklo* behaves like a liquid-context jer, not like a genuine reduced vowel.

4. **Cross-IE and Slavic-internal divergence.** Every IE branch independently inserts a *different* vowel at PIE syllabic resonant positions. Within Slavic, five branches produce five different vowels at the same position (Czech *vlk*, Serbian *vuk*, Russian волк, Polish *wilk*, Bulgarian вълк). If a real vowel had been inherited, the question arises why five branches would each independently replace it. Independent epenthesis into an empty position requires no such explanation.

5. **The jer-color diagnostic.** If liquid-context jers were real vowels, their front/back quality should reflect Slavic-internal phonological conditioning. Instead, it reflects PIE etymology with near-categorical accuracy (Table 8: 94.6% of ь+liquid words trace to PIE syllabic resonants; 0% of ъ+liquid words do). This distribution is expected from a notation preserving inherited distinctions, not from a phonologically active segment whose quality is determined by its synchronic environment.

The cumulative weight of these arguments favors the stronger claim: liquid-context jers were not merely inert vowels but notational representations of syllabic consonants that never had independent vocalic status in Slavic.

7.2 A reformulated Havlík's law

Under the standard formulation, Havlík's law requires an implicit exception: "Strong jers vocalize—*except in liquid environments, where the liquid becomes syllabic instead.*" Under our reformulation, no exception is needed:

> **Havlík's law (reformulated):** In Proto-Slavic words containing genuine reduced vowels (ъ from **u*, ь from **i*), weak-position jers are deleted and strong-position jers vocalize to full vowels. Orthographic jers representing syllabic consonants (from PIE **l̥*, **r̥*) are outside the domain of the law.

This reformulation makes Havlík's law **more regular**, not less. The apparent exception for liquid contexts vanishes because the jers in those positions were never part of the law's input.

7.3 Implications for Proto-Balto-Slavic

If liquid-context jers were not real vowels, the standard PBS reconstruction of syllabic resonants via an intermediate **iR*/**uR* stage requires re-examination. Under the standard model, Slavic first epenthesized a vowel (PIE **wl̥kʷos* → PBS **wilkas*) and then reduced it back to a syllabic consonant—two changes that cancel each other out. Our data raise the question of whether the intermediate full-vowel stage existed in Slavic at all. We defer a full treatment of this question to a separate study; the Havlík's law asymmetry documented here is one empirical argument that such a study must address.

8 Open questions

We flag two questions that require further research:

1. **Non-PIE-syllabic-resonant liquid jers across Slavic.** The back-jer (ъ) + liquid words identified in Table 8 include loanwords (**dъlgъ* 'debt' from Gothic, **xъlmъ* 'hill' and **pъlkъ* 'regiment' from Germanic), laryngeal sequences (**kъrkъ* 'neck', **gъrdlo* 'throat'), and substrate words. These do not derive from PIE syllabic resonants, yet their Czech outcomes pattern identically with Category B: *dluh*, *chlum*, *pluk* (hard l → /u/); *krk*, *hrdlo* (syllabic r). None shows /e/ vocalization. This suggests that the Category A/B split tracks *phonological context* (liquid adjacency), not just PIE etymological class. A fuller investigation of these cases across Serbian and Slovak would further test this generalization.

2. **Old Russian manuscripts.** Is there evidence for a syllabic consonant stage in Old Russian *before* full vocalization? If East Slavic manuscripts show variation between jer + liquid and liquid-only spellings (as OCS manuscripts do), this would support a syllabic consonant intermediate stage.

9 Conclusion

We have documented a categorical asymmetry in the outcomes of Havlík's law across Slavic: strong-position jers vocalize to full vowels in 100% of non-liquid cases—Czech /e/ ($n = 23$), Serbian /a/ ($n = 22$), Slovak /e/ ($n = 23$)—but in 0% of liquid-adjacent cases ($n = 27$). Across three independent branches, 68 out of 68 non-liquid strong jers vocalized while zero liquid-adjacent strong jers did. The seven Modern Czech words with /e/ near liquids are secondary epenthesis, documented in Old Czech as syllabic consonant forms, and separated from Havlík's law by two to three centuries. The resolution of hard syllabic /l/ to /u/ (not /e/) confirms a process distinct from jer vocalization. In Russian, the vowel quality at liquid-jer positions follows polnoglasie patterns (front jer ь → /o/ before /l/, matching **TelT* → ToloT) rather than standard jer vocalization, indicating that East Slavic "vocalization" was independent epenthesis to resolve syllabic consonants, not Havlík's law.

We argue that this asymmetry constitutes evidence that liquid-context jers in OCS were notational conventions for syllabic consonants, not genuine reduced vowels. Havlík's law, reformulated as applying only to real reduced vowels (from PIE **u*/**i*), becomes more regular: the "liquid exception" disappears because there was never a vowel to except.

The argument has implications for Proto-Balto-Slavic reconstruction: if liquid-context jers were never real vowels, the standard PBS *iR/ *uR intermediate stage requires re-examination—a question we defer to a separate study.

Our proposal does not invalidate Havlík's law. It refines its domain: the law correctly describes the behavior of genuine reduced vowels, which is exactly the category it was designed to explain. What it does not describe is the behavior of positions that never had a vowel to begin with.

A final observation underscores the point. PIE *$*w\textit{l̥}k^{w}os$* 'wolf' yields Czech *vlk*, Serbian *vuk*, Russian волк, Polish *wilk*, and Bulgarian вълк—five branches, five different vowels at the same position. Under the standard model, a single inherited reduced vowel produced five different outcomes; under ours, each branch independently resolved a syllabic consonant, just as each IE branch did before them (Sanskrit *r̥*, Greek /a/, Latin /o/, Germanic /u/, Lithuanian /i/; cf. Mayer 1991). The instability is inherent to the phonological position, not to the particular vowel that happened to occupy it.

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